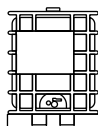


Product	Sinopec Type U Insulated Transformer Oil
Summary	Product description Sinopec Type U Insulated Transformer Oil is an uninhibited transformer oil. Developed and formulated to deliver solid resistance to oil degradation, Type U Insulated transformer oil provides good oxidation stability thanks to its natural inhibitors. This increases the possibilities for a longer transformer life with less maintenance. They are designed for insulating and cooling transformers, and other electrical insulating applications. There are two kinds of transformer oil, U-10°C transformer oil and U-40°C transformer oil, which meet the requirements of GB2536-2011(IEC60296-2003). Sinopec Type U Insulated transformer oil doesn't contain polychlorinated biphenyls (PCBs).

Available sizes



DRUM – 200L



IBC – 1000L

Also
available
in bulk

Applications

Sinopec Type U Insulated Transformer Oil is suitable for use in:

- Oil-filled transformers, where it acts as a non-conducting insulating and cooling medium.
- Other electrical insulating applications.

Features and benefits

- Excellent dielectric (insulating) properties, effectively prevent discharge in an electric field.
- Low viscosity and low density lead to the oils having good cooling properties and good heat transfer characteristics at room temperature and below, and so ensure effective cooling of transformer core and coils.
- Even at very low temperatures the oil viscosity remains low, and therefore allows the transformer to be started up under very cold conditions without problem.
- Excellent oxidation stability characteristics are provided by the stable base oil together with the high-performance oxidation inhibitor system, and prevent the formation of acidic material and sludge in service, extending the life of the oil.

Sinopec Type U transformer oil meets the requirements of GB2536-2011(IEC60296-2003):

- Not inhibitors transformer oil , the oil complies with GB2536-2011(IEC60296-2003).
- Type U transformer oil is rigorously analysed and passes the following corrosion tests:
 - SH/T0804
 - IEC 62535
 - DIN 51353

Product Data Sheet

Typical Data

Grade	SPECIFICATION LIMITS of GB2536- 2011(IEC60296-2003)	Type U Insulated Transformer Oil		Reference method
		U-10℃	U-40℃	
Kinematic viscosity, mm ² /s				
cSt @ 40℃, GB/T 265	≤12	9.949	9.530	ISO 3104
cSt @ -10℃, GB/T 265	≤1800	107.2	—	ISO 3104
cSt @ -40℃, NB/SH/T 0837	≤2500	—	1365	IEC 61868
Pour point, °C, GB/T 265	*	-39	-54	ISO 3016
Water Content, mg/kg, GB/T 7600	< 30	16	19	IEC 60814
Dielectric Breakdown (as received) , kV, GB/T 507	30	54.0	48.0	IEC 60156
Dielectric dissipation factor @ 90 °C , GB/T5654	0.005	0.0006	0.0009	IEC 60247 or IEC 61620
Density @ 20 °C , kg/m ³ , GB/T 1884&GB/T 1885	0.895	833.6	831.0	ISO 3675 or ISO 12185
Appearance, vision	Clear & bright	Clear & bright	Clear & bright	vision
Acidity, mg KOH/g, NB/SH/T 0836	≤0.01	<0.01	<0.01	IEC 62021-1 or 62021-2
Interfacial tension, mN/m, GB/T 6541	40	51.4	55	EN 14210 or ASTM D971
Total Sulfur content, %, SH/T 0689	No requirement	0.0018	0.0015	IP 373 or ISO 14596
Corrosive sulfur, SH/T0804	Not corrosive	Not corrosive	Not corrosive	DIN 51353
Inhibitors ,%, SH/T0802	Not detectable	Not detectable	Not detectable	IEC 60666
2-Furfural & related compounds, mg/kg, NB/SH/T 0812	≤0.1	<0.05	<0.05	IEC 61198
Oxidation stability, 164 h @ 120℃, NB/SH/T 0811				
total acidity, mg KOH/g	≤1.2	0.20	0.47	IEC 61125:1992 (Method C)
sludge, wt%	≤0.8	0.01	<0.01	
DDF at 90℃ , GB/T5654	≤0.500	0.0280	0.0052	
Flash point (PMCC), °C, GB/T 261	135	165.0	154.0	ISO 2719
PCA Content , %, NB/SH/T 0838	≤ 3	0.27	0.70	IP 346
PCB Content, mg/kg , NB/SH/T 0803	Not detectable (< 2 mg/kg)	Not detectable	Not detectable	IEC 61619
Note: * Pour point , °C U-10℃ Transformer oil ≤-20 U-40℃ Transformer oil ≤-50				

These data are given as an indication of typical values and not as exact specifications.

Industry and OEM specifications

Sinopec Type U Super Insulated Transformer Oil meets the performance requirements of the following industry specifications:	
GB	2536-2011
IEC	60296-2003

Product Data Sheet

Accuracy of information

Data provided in this PDS is typical and subject to change as a result of continuing product research and development. The information given was correct at the time of printing. The typical values given are subject to variations in the testing procedures and the manufacturing process may also result in slight variations. Sinopec guarantees that its lubricants meet any industry and OEM specifications referred to on this data sheet.

Sinopec cannot be held responsible for any deterioration in the product due to incorrect storage or handling. Information on best practice is available from your local distributor.

Product and environmental safety

This product should not cause any health problems when used in the applications suggested and when the guidance provided in the Material Safety Data Sheet (MSDS) is followed. Please consult the MSDS for more detailed advice on handling; MSDSs are available from your local distributor. Do not use the product in applications other than those suggested.

As with all products, please take care to avoid environmental contamination when disposing of this product. Used oil should be sent for reclamation/recycling or, if not possible, must be disposed of according to relevant government/authority regulations.

The SINOPEC trademark is registered and protected.

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Sinopec Type U Insulated Transformer Oil